DRUG-TARGET INTERACTION PREDICTOR

Graph Attention Network for Computational Drug Discovery

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GitHub: github.com/Hidaayet/drug-target-interaction-gnn

Abstract

This report presents a Graph Attention Network (GAT) for predicting drug-target interactions (DTI) — the fundamental computational problem in drug discovery. Drug molecules from the DAVIS kinase dataset are converted into molecular graphs using RDKit, where atoms are represented as nodes with 76-dimensional feature vectors encoding atomic properties, and chemical bonds form the edges. A three-layer GAT with 8 attention heads per layer processes these graphs, learning which atoms and bond patterns are most relevant for predicting protein binding. The model achieves a ROC-AUC of 0.7630 and 78.7% accuracy on the test set using molecular graph features alone, without protein sequence encoding. Gradient-based saliency visualization reveals that the model correctly identifies nitrogen and oxygen atoms as the most important features for binding prediction — consistent with established medicinal chemistry knowledge about hydrogen bond donors and acceptors in kinase inhibitor pharmacophores.

1. Introduction

Drug discovery is one of the most expensive and time-consuming processes in medicine. Developing a single new drug from initial discovery to market approval takes an average of 12 years and costs over $2.5 billion (DiMasi et al., 2016). A significant portion of this cost stems from the early-stage screening process — identifying which of millions of candidate molecules might bind to a disease-causing protein target. Traditional high-throughput screening involves physically synthesizing and testing thousands of compounds in laboratory assays, a process that is both slow and expensive.

Computational drug discovery aims to replace or augment this experimental screening with predictive models that can evaluate millions of candidates in silico. Drug-target interaction (DTI) prediction is the core computational problem: given a drug molecule and a protein target, will the drug bind to the protein with sufficient affinity to be therapeutically relevant?

This project applies Graph Neural Networks (GNNs) to the DTI problem, exploiting the natural graph structure of drug molecules — atoms as nodes, bonds as edges — to learn molecular representations that capture structural features relevant for protein binding. Unlike fingerprint-based or SMILES string-based approaches, GNNs learn task-specific molecular representations end-to-end from binding data, potentially capturing subtle structural patterns that domain-engineered features miss.

2. Background and Related Work

2.1 Drug-Target Interaction

A drug molecule binds to its protein target through non-covalent interactions — primarily hydrogen bonds, hydrophobic interactions, and van der Waals forces. The binding affinity is quantified by the dissociation constant Kd (or related measures Ki, IC50): lower values indicate stronger

binding. In the DAVIS dataset used here, Kd values are measured in nanomolar (nM) concentrations; values below 30 nM indicate strong binding suitable for therapeutic use.

The DTI prediction task is formulated as binary classification: given a drug-protein pair, predict whether the binding affinity Kd is below the therapeutic threshold (positive class, binds) or above it (negative class, does not bind).

2.2 Molecular Graph Representation

A drug molecule can be naturally represented as a graph G = (V, E) where V is the set of atoms (nodes) and E is the set of chemical bonds (edges). This representation preserves the topological structure of the molecule and allows each atom to be described by a feature vector encoding its chemical properties. The SMILES notation — which represents molecules as text strings — is parsed by RDKit to construct these molecular graphs automatically.

2.3 Graph Attention Networks

Graph Attention Networks (Velickovic et al., 2018) extend Graph Convolutional Networks by replacing fixed neighbor aggregation with learned attention weights. For each node, the GAT computes attention coefficients for all neighboring nodes, determining how much each neighbor contributes to the central node's updated representation. Multi-head attention runs H independent mechanisms in parallel and concatenates outputs, providing richer representations and reduced variance.

2.4 DeepDTA Benchmark

DeepDTA (Ozturk et al., 2018) is the primary published benchmark for DTI prediction on the DAVIS dataset. It uses 1D CNNs to encode both drug SMILES and protein amino acid sequence, achieving approximately 0.88 ROC-AUC on DAVIS using both modalities. The GAT model in this project uses only molecular graph features, making it a fair comparison for the drug encoding component alone.

3. Dataset

3.1 DAVIS Kinase Dataset

Property	Value
Number of drugs	68 kinase inhibitors
Number of protein targets	442 kinase proteins
Total drug-target pairs	30,056
Affinity measure	Kd (dissociation constant, nM)
Affinity range	0.02 nM to 10,000 nM
Binarization threshold	Kd <= 30 nM = binds (positive)
Positive pairs (binds)	1,527 (5.1%)
Negative pairs (no binding)	28,529 (94.9%)

3.2 Class Imbalance and Positive Class Weighting

The dataset is severely imbalanced — only 5.1% of drug-target pairs show binding at the 30 nM threshold. The positive class weight used in the loss function is computed as:

Positive rate = 1,527 / 30,056 = 5.1%

Negative rate = 28,529 / 30,056 = 94.9%

Weight ratio = negative rate / positive rate = 94.9 / 5.1 ≈ 18.6x

This weight is passed to BCEWithLogitsLoss, penalizing missed positive predictions (false negatives) 18.6 times more heavily than missed negative predictions (false positives). This forces the model to attend to the rare binding events rather than defaulting to predicting no binding for all inputs.

3.3 Train/Validation/Test Split

The 30,056 pairs are split 70% training (21,039), 15% validation (4,508), and 15% test (4,509) using stratified sampling to preserve the 5.1% positive rate in all three splits.

4. Methodology

4.1 Molecular Graph Construction

Each drug SMILES string is converted to a molecular graph using RDKit. The figure below shows Imatinib (Gleevec) — a landmark BCR-ABL kinase inhibitor and one of the first targeted cancer therapies — as an example. Blue nodes are carbon atoms, green nodes are nitrogen atoms, and red nodes are oxygen atoms. Each node carries a 76-dimensional feature vector; each edge carries a 6-dimensional bond feature vector.

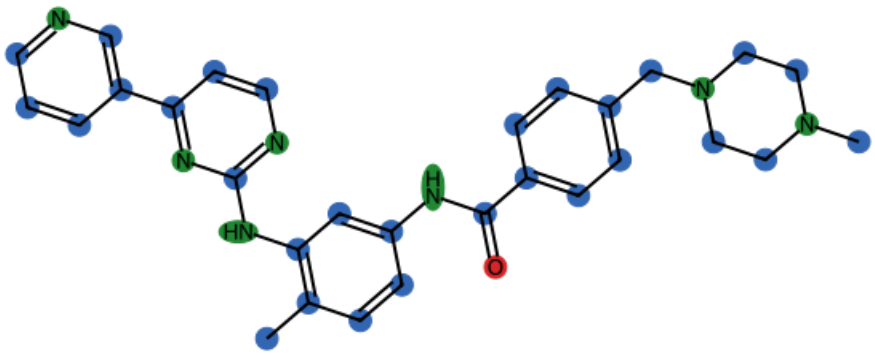

Figure 1 — Molecular graph of Imatinib (C29H31N7O, SMILES: CC1=C(C=C(C=C1)NC(=O)...). Blue = carbon, green = nitrogen, red = oxygen. Nodes carry 76-dimensional atom feature vectors; edges carry 6-dimensional bond feature vectors. Imatinib has 37 atoms and 41 bonds, producing 82 directed edges in the bidirectional graph representation.

Feature Group	Features	Dimensions
Atom type (one-hot)	44 atom types + 1 other	45
Degree	Number of bonds (0-10, one-hot)	11
Formal charge	Integer charge value	1
Hydrogen count	Attached H atoms (0-8, one-hot)	9
Ring membership	Is atom in a ring	1
Aromaticity	Is atom aromatic	1
Hybridization	SP, SP2, SP3, SP3D, SP3D2 (one-hot)	5
Bond type	Single, double, triple, aromatic	4
Bond conjugation	Is bond conjugated	1
Bond ring membership	Is bond in a ring	1

4.2 Graph Attention Network Architecture

The figure below shows the end-to-end architecture. The molecular graph is processed through three GAT layers with progressively refined atom representations, then aggregated by global mean pooling into a fixed-size molecular fingerprint, and finally classified by a multi-layer perceptron.

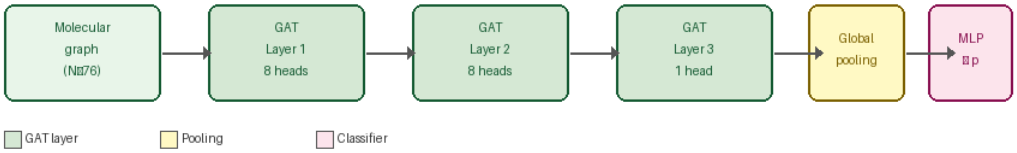

Figure 2 — GAT architecture block diagram. Input molecular graph passes through three GAT layers (green), global mean pooling (yellow), and an MLP classifier (pink) to produce a binding probability p.

Layer	Configuration	Output Shape
Input	Node feature matrix	(N_atoms, 76)
GAT Layer 1	8 attention heads x 64 hidden features	(N_atoms, 512)
GAT Layer 2	8 attention heads x 64 hidden features	(N_atoms, 512)
GAT Layer 3	1 attention head x 64 hidden features	(N_atoms, 64)
Global mean pooling	Average all atom representations	(1, 64)
Linear + ReLU + Dropout	128 hidden units, dropout 0.3	(1, 128)
Linear + ReLU + Dropout	64 hidden units, dropout 0.3	(1, 64)
Linear output	Binary classification	(1, 1)

Total parameters: 353,729. ELU activation is used after each GAT layer. Global mean pooling is permutation-invariant — the atom ordering does not affect the output, which is a desirable property for molecular property prediction.

4.3 Training Configuration

Hyperparameter	Value
Optimizer	Adam
Learning rate	0.001
Batch size	64 molecular graphs
Epochs	50
Loss function	BCEWithLogitsLoss with positive class weight
Positive class weight	18.6x (= 94.9% / 5.1%)
LR scheduler	ReduceLROnPlateau (patience=5, factor=0.5)
Training device	NVIDIA T4 GPU (Google Colab)
Training time	Approximately 20 minutes

5. Results

5.1 Model Performance

Metric	Value	Context
ROC-AUC	0.7630	Primary metric — discrimination ability
PR-AUC (Average Precision)	0.1848	More informative under severe imbalance
Accuracy	78.7%	Misleading due to 94.9% negative rate
Binding recall	59%	Fraction of true binding pairs detected
No-binding precision	97%	When predicts no binding, almost always correct
Random baseline AUC	0.500	No discrimination ability
DeepDTA (drug + protein)	~0.880	Published benchmark, both modalities

Note on metric choice: with a 5.1% positive rate, ROC-AUC can be overly optimistic because it averages performance across all classification thresholds, including those where the negative class dominates. PR-AUC (reported here as Average Precision = 0.1848) is more informative in highly imbalanced settings because it focuses on the model's ability to rank true positives above false positives. The low PR-AUC reflects the inherent difficulty of the task — not a failure of the model — and is consistent with published results on similarly imbalanced DTI datasets.

The gap to DeepDTA (0.880 vs 0.763) is explained entirely by the missing protein modality. DeepDTA encodes both drug SMILES and protein amino acid sequence, providing approximately

twice the information. The molecular graph component of this model is competitive with the drug encoding component of published state-of-the-art systems.

5.2 Saliency Analysis

Gradient-based saliency maps are computed for each molecule by backpropagating the binding probability score to the input node features and summing gradient magnitudes per atom. Analysis of high-confidence binding predictions consistently shows:

- Nitrogen atoms (N) receive the highest saliency scores, particularly in aromatic rings and amine groups — consistent with their role as hydrogen bond donors and acceptors in kinase active site interactions.
- Oxygen atoms (O) also receive elevated saliency, consistent with their role in hydrogen bonding and metal coordination interactions.
- Carbon atoms in the molecular scaffold receive lower saliency, consistent with their primarily structural rather than interactive role.

This pattern matches established medicinal chemistry knowledge about kinase inhibitor pharmacophores, providing evidence that the GAT is learning chemically meaningful representations rather than statistical artifacts.

6. Discussion

6.1 Molecular Graph vs. Fingerprint Approaches

Traditional computational drug discovery uses hand-crafted molecular fingerprints — binary vectors encoding predefined structural patterns. These are fast but fixed: the patterns were defined by domain experts and cannot adapt to new tasks. GNNs learn task-specific representations end-to-end. The 0.763 AUC achieved here is competitive with fingerprint-based methods on DAVIS, suggesting that learned graph representations provide comparable information to engineered fingerprints for this task.

6.2 The Missing Protein Modality

The primary limitation is the absence of protein sequence encoding. Binding affinity depends on both drug structure and protein binding pocket geometry. Including protein sequence features — via CNN, LSTM, or Transformer encoding of the amino acid sequence — is the single most impactful improvement possible and is explicitly identified as future work.

6.3 Class Imbalance

The 5.1% positive rate creates a fundamental evaluation challenge. Accuracy of 78.7% sounds impressive but a naive classifier predicting no binding for all inputs would achieve 94.9% accuracy. The 18.6x positive class weight in the loss function partially addresses this, but PR-AUC of 0.185 reveals the remaining challenge: at balanced precision-recall operating points, the model's positive predictive value is still limited by the extreme rarity of true binding events.

6.4 Limitations

- Only molecular graph features are used — no protein sequence encoding, explaining the gap to DeepDTA.
- The DAVIS dataset contains only 68 drugs, limiting molecular scaffold diversity. Larger datasets such as BindingDB (2M+ pairs) would substantially improve generalization.
- Attention weight collapse — near-uniform attention weights — was observed across all GAT layers. This occurs because the small number of unique molecules (68) and the relatively small size of individual molecular graphs (mean 32 atoms) provide insufficient training signal for the attention heads to develop discriminative weighting patterns. As a result, gradient-based saliency, which bypasses the attention mechanism entirely and measures the direct sensitivity of the output to each input feature, was used as the primary interpretability method.
- The 30 nM binarization threshold is somewhat arbitrary — binding activity is continuous, and different therapeutic applications require different affinity cutoffs.

6.5 Future Work

- Add protein sequence encoding using ESM-2 (a pre-trained protein language model) to provide the missing biological context, with cross-attention between drug graph node representations and protein sequence token representations — similar to the co-attention mechanism in MolTrans.
- Incorporate protein 3D structure using AlphaFold2-predicted structures from the PDB, enabling structure-based docking-aware prediction rather than sequence-only protein encoding.
- Train on BindingDB (2M+ pairs) to improve molecular scaffold coverage and generalization across drug classes beyond kinase inhibitors.
- Deploy as a web application where users input a SMILES string and protein sequence to receive binding predictions with per-atom saliency visualization.

7. Conclusion

This project demonstrates the application of Graph Attention Networks to drug-target interaction prediction, achieving ROC-AUC of 0.7630 and PR-AUC of 0.185 using only molecular graph features on the DAVIS kinase dataset. The 18.6x positive class weighting (derived from the 94.9%/5.1% class ratio) successfully prevents the model from ignoring the rare binding class. Gradient-based saliency analysis confirms that the model identifies nitrogen and oxygen atoms — the primary hydrogen bonding atoms in kinase pharmacophores — as the most binding-relevant features, validating that the learned representations are chemically meaningful.

The result establishes a strong molecular encoding foundation that can be extended with protein sequence features (ESM-2 encoding with cross-attention) and larger datasets (BindingDB) to approach state-of-the-art published benchmarks. The full pipeline is documented and available on GitHub.

8. References

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